SUPERSTORE SALES ANALYSIS

Graduation Project - DEPI Data Analysis Track

Team size	Dataset	Business requirements
5 members	**Superstore Sales**	**40 questions**
Main tool	**Final deadline**	**Final demo**
Power BI	**4 Dec 2026**	**Demo video included**

Project goal

Use Power BI to clean and analyze the Superstore sales data, answer 40 business questions, and build a clear interactive dashboard. The results will help explain sales, customers, products, orders, time trends, and shipping performance.

Tools we will use

- **Power BI Desktop** — data model, measures, charts, and dashboard.
- **Power Query** — data cleaning and preparation inside Power BI.
- **DAX** — calculations and KPIs.
- **GitHub** — store project files, documentation, and final deliverables.

Note: We will not use SQL or Python. Data cleaning and analysis will be done with Power Query, Power BI, and DAX.

1. Project Plan and Deadlines

All project documents and final files should be uploaded to the team GitHub repository.

Phase	What we will finish	DEPI deadline
1. Planning & Requirements	Project proposal, team roles, GitHub setup, and 40 business questions.	**16 Oct 2026**
2. Analysis & Data Design	Literature review, data dictionary, schema/data model, and cleaning plan.	**6 Nov 2026**
3. Implementation	Clean data, create DAX measures, and build the Power BI dashboard.	**30 Nov 2026**
4. Final Testing & Presentation	Test dashboard, finish report and slides, record demo video, and prepare final delivery.	**4 Dec 2026**

Simple weekly plan

- **Week 1:** Clean the dataset in Power Query and document the changes.
- **Week 2:** Review the columns and agree on the 40 questions the data can answer.
- **Week 3:** Build the Power BI data model, DAX measures, and dashboard pages.
- **Week 4:** Test the results, write the final report, prepare slides, and record the demo video.

Final deliverables

- Cleaned dataset / Power BI-ready data and a short cleaning log.
- Data dictionary and data model/schema diagram.
- Power BI dashboard with charts, filters, and the required DAX measures.
- Final report with key findings and simple recommendations.
- Final presentation slides.
- Short video: a short recording showing the dashboard, filters, and main findings.
- GitHub repository containing the project files and documentation.

2. Team Members and Responsibilities

Team member	Main responsibilities
Maryam Mohammed (Team Lead)	Manage the project and deadlines; create and organize the GitHub repository; write the project proposal and risk plan; track tasks; create half of the required DAX measures; take part in cleaning the dataset; check that all files are uploaded.
Habiba Amin	Clean and prepare the dataset using Power Query; keep the data cleaning log; help validate cleaned data; write and organize the final report.
Mohammed Yasser	Prepare the data schema/model in Power BI; create the data dictionary; explain tables, columns, data types, and relationships; check that the model relationships work correctly.
Mohammed Samir	Plan the dashboard layout and visual design; choose suitable charts and filters; build the dashboard pages in Power BI; check that the dashboard is easy to read.
Razan Mohammed	Create the remaining DAX measures; check calculations with the team; prepare the final presentation slides; present the main findings and recommendations.

Shared tasks

- Everyone reviews the 40 questions and confirms that the dataset has the needed columns.
- Everyone tests the dashboard and checks that totals match the cleaned data.
- The team lead checks progress, but each member is responsible for finishing their own tasks.
- Maryam and Habiba coordinate the final report; Razan leads the slides; Mohammed Samir records the demo video with help from the team.

Dataset Overview

The project uses the Superstore Sales dataset. Each row is one product line inside an order, so one order can appear in several rows. The dataset has **16 source columns** and contains sales values only: there are **no Profit, Discount, or Quantity columns**, and no outside data is added to it.

Rows	Orders	Customers	Products	Period
9,800	**4,922**	**793**	**1,861**	**3 Jan 2015 - 30 Dec 2018**

Row count includes one fully repeated row (Order US-2015-150119, sales 281.37). It is kept because the dataset has no Quantity or Row ID that would prove it is an error, so it may be a legitimate repeated line. Orders = distinct Order ID, customers = distinct Customer ID, products = distinct Product ID.

Source columns

Column	Data type	Meaning
Order ID	Text	Identifier of the order.
Order Date new	Date	Date the order was placed.
Ship Date New	Date	Date the order was shipped.
Ship Mode	Text	Shipping method: Standard Class, Second Class, First Class, or Same Day.
Customer ID	Text	Identifier of the customer.
Customer Name	Text	Name of the customer.
Segment	Text	Customer segment: Consumer, Corporate, or Home Office.
City	Text	City of the customer.
State	Text	State of the customer.
Postal Code	Text	Postal code. 11 values are missing and are not filled from outside data.
Region	Text	Sales region: Central, East, South, or West.
Product ID	Text	Identifier of the product. It is the product key used in all rankings and counts.
Category	Text	Product category: Furniture, Office Supplies, or Technology.
Sub-Category	Text	Product sub-category.
Product Name	Text	Name of the product. 32 Product IDs carry two different names, and 16 generic names (for example Staples) are shared by several Product IDs.
Sales	Decimal number	Sales amount of the row.

Columns added in Power BI

Column	Data type	Meaning
Shipping Days	Whole number	Days between the order date and the ship date.
Shipping Speed	Text	Fast (0-2 days), Normal (3-4 days), or Slow (5+ days).
Order Total	Decimal number	Total sales of the whole order.
Order Range	Text	Order value range: 1) Under 100, 2) 100 - 500, 3) 500 - 1000, 4) Over 1000.
Year	Whole number	Year of the order date.

Definitions used in the analysis

- **Order:** one distinct Order ID. **Sales record:** one row of the dataset.
- **Product:** one Product ID. Rankings and counts use Product ID; Product Name is shown as the label next to it.
- **Shipping time:** number of days between Order Date and Ship Date.

3. The 40 Business Requirements

These are the questions the dashboard should answer. Every question below can be answered from the available dataset columns.

A. Sales Performance

1. What is the total sales revenue?
2. How many orders did the business receive in total?
3. Which month had the highest sales?
4. Which month had the lowest sales?
5. What is the average sales value per order?
6. How do sales change from one year to the next?
7. Which 10 products have the highest total sales?
8. How many orders are in each sales-value range?

B. Customer Analysis

9. How many unique customers are in the dataset?
10. Which 10 customers have the highest total sales?
11. Which customer segment has the most orders?
12. Which customer segment brings in the most sales?
13. What is the average order value for each customer segment?
14. Which customers placed the most orders?
15. What share of total sales comes from each customer segment?
16. Which states have the most sales from the top 50 customers?

C. Product Analysis

17. Which product category has the highest sales?
18. Which product category has the most orders?
19. Which five Office Supplies sub-categories have the highest sales?
20. Which sub-category has the lowest sales?
21. Which product appears in the most sales records?
22. What percentage of total sales comes from each product category?
23. How do category sales compare across customer segments?
24. What is the average sales value per record for Technology products?

D. Geographic and Shipping Analysis

25. Which region has the highest total sales?
26. Which five states have the highest sales?
27. Which cities have the highest sales?
28. Which state has the lowest total sales?
29. Which shipping mode is used most often?
30. How many orders use each shipping mode?
31. What is the average order value for each shipping mode?
32. What is the average shipping time for each region?

E. Order Volume, Time Trends and Shipping Speed

33. What is the year-over-year sales growth percentage for each year?
34. Which product sub-category has the most orders?
35. Which 10 products have the lowest total sales?
36. Which region has the most orders?
37. Which quarter of the year has the highest sales?
38. What is the average shipping time for each shipping mode?
39. What share of total sales comes from the top 10 customers?
40. How do sales for each customer segment change from year to year?

4. How We Will Build the Project

Step 1 — Prepare the data

- Open the dataset in Power BI and use Power Query to check missing values, duplicates, incorrect data types, and date formats.
- Keep a short cleaning log that lists what was changed and why. Missing Postal Code values are kept as they are (or marked "Unknown") because no outside data is allowed and no question uses this column.
- The cleaning log should record: Row ID and Country removed; text dates converted to dates (dd/MM/yyyy); the missing leading zero added to 429 Postal Codes; the 11 missing Postal Codes marked "unknown"; and the one fully repeated row kept, with the reason given in the Dataset Overview.

Step 2 — Build the data model

- Write down each column name, meaning, and data type in the data dictionary.
- Create a clear model/schema in Power BI and check the relationships between tables, where needed.

Step 3 — Create DAX measures

- Create reusable measures for total sales, order count, unique customers, average order value, average sales per customer, average shipping days, year-over-year growth, share of total sales, and other needed calculations.
- Maryam creates about half of the measures; Razan creates the rest. They review each other's results.

Step 4 — Build the dashboard

- Use clear charts, cards, slicers, and titles. Keep the layout simple and consistent.
- Group the dashboard into pages such as Sales, Customers, Products, Region & Shipping, and Orders & Trends.

Step 5 — Test and finish

- Check key totals and calculations against the dataset.
- Test slicers, page navigation, and chart interactions.
- Write the report and recommendations, prepare the presentation, and record the demo video.

Suggested dashboard KPIs

Total Sales	**Total Orders**	**Unique Customers**
Average Order Value	**Average Shipping Days**	**Average Sales per Customer**

Final checklist

- [ ] 40 business questions reviewed and approved.
- [ ] Dataset cleaned in Power Query and cleaning log saved.
- [ ] Data dictionary and data model/schema completed.
- [ ] DAX measures checked.
- [ ] All dashboard pages and filters tested.
- [ ] Final report completed.
- [ ] Final presentation slides completed.
- [ ] Demo video recorded and included in final deliverables.
- [ ] All documents and project files uploaded to GitHub.

5. Risk Assessment and Project KPIs

This page lists the main project risks with a simple plan to reduce each one, and the indicators we will use to check that the project is successful.

Risk assessment and mitigation plan

Risk	Likelihood	Impact	Mitigation
Dashboard numbers do not match the cleaned data	Medium	High	Compare every key number with a reference table before delivery; each member tests the totals.
Year-over-year growth or Top N visuals give wrong results when filters are used	Medium	High	Test every slicer; build the year-over-year measure so it works only when one year is selected.
Power BI file is lost or members overwrite each other's work	Medium	High	Upload a version to GitHub after each phase; only one member edits the model at a time.
The CSV path is tied to one computer, so Refresh fails for other members	High	Medium	Use a file-path parameter, or keep the CSV inside the repository.
Wrong interpretation of the data (for example adding order counts across categories)	Medium	Medium	Write the definitions (order, product, shipping time) in the report and explain the counts.
Not enough time before the 4 Dec 2026 final deadline	Medium	High	Prepare slides and demo video in parallel with building the dashboard; assign tasks clearly.
A team member is late with a task	Medium	High	Weekly progress check by the team lead; members help each other when someone falls behind.
Limited data: no Profit, Discount, or Quantity columns	Fixed	Medium	State the scope clearly in the report; keep recommendations about sales, orders, and shipping only.

Project KPIs (measures of project success)

Project KPI	Target
Business questions answered in the dashboard	40 of 40
Dashboard numbers matching the cleaned data	100%
DAX measures checked by another team member	All measures
Dashboard pages tested (slicers, navigation, interactions)	All pages
Deliverables uploaded to GitHub	All, before 4 Dec 2026
Phase deadlines met (16 Oct, 6 Nov, 30 Nov, 4 Dec)	4 of 4

6. Problem Statement, Objectives and Scope

Problem Statement

The Superstore data has 9,800 sales rows (4,922 orders from 793 customers) covering 2015 to 2018. As it comes, the dates are stored as text, some postal codes are missing, and there's nothing that shows the numbers in a simple way. So it's hard to tell how sales move over the years, who the best customers are, which products and regions sell the most, and how long shipping takes.

Project Objectives

Our goal is to clean this data in Power BI and use it to answer 40 business questions about sales, customers, products, regions, shipping and time trends. We will build an interactive dashboard with five pages (Sales, Customers, Products, Region & Shipping, Orders & Trends), then write a report with the main findings and some simple recommendations, plus the slides and a demo video.

Project Scope

We only use the Superstore dataset as it is, with no outside data. It has sales values only, so there's no profit, discount or quantity analysis. Everything is done with Power BI, Power Query and DAX (no SQL or Python).